

Weekly CT - 04 | Question Paper

Cambridge International AS and A Level Computer Science 9618

Data representation | Networks | Hardware and memory

Time: 40 minutes		Total : 40
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Instructions: Answer all questions in the spaces provided. Show working for calculations. Marks for each part are shown in brackets. Questions 7 and 8 are bonus items.

1. Data representation [7]

(a) Convert the unsigned binary number [1]

11010110 to denary.

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(b) Convert the hexadecimal number 9E to [2]

denary. Show your working.

Working:

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(c) Represent -37 as an 8-bit two's complement [2]

binary number. Show your working.

Working:

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(d) State one reason hexadecimal is commonly **[1]**

used when representing binary data.

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.....

(e) State the number of characters that can be **[1]**

represented using standard ASCII.

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2. Multimedia representation and **[7]**

compression

(a) A bitmap image has a resolution of 1024 x 512 **[3]**

pixels and a colour depth of 24 bits per pixel.

Calculate the uncompressed file size in mebibytes

(MiB). Show your working.

Working:

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(b) Define sample rate in digital sound. **[1]**

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(c) Explain two effects of increasing the sample

[2]

rate when recording the same sound.

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.....

(d) Name one lossless compression method that

[1]

can be used for a bitmap image.

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3. Networks and network devices

[8]

(a) Describe two features of a peer-to-peer network.

[2]

Complete the table by naming the network device described in each row.

[4]

Description	Device
Forwards packets between different IP networks using destination IP addresses.	
Connects devices within a LAN and forwards frames to the correct port using MAC addresses.	
Converts/modulates and demodulates signals so a network can communicate over an ISP access line.	
Sends incoming data out to all other ports rather than selecting one destination port.	

(c) A student mainly uses a desktop computer for streaming high-resolution video and playing online games. Recommend a wired or wireless connection to the router and justify your choice.

[2]

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4. Internet, IP addressing and cloud

[8]

computing

(a) Explain one difference between a public IP

[2]

address and a private IP address.

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(b) Explain the difference between a static IP

[2]

address and a dynamic IP address.

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(c) A device has the IPv4 address

[2]

192.168.10.70/26. State the subnet mask and the

network address.

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(d) State one benefit and one drawback of using

[2]

cloud storage.

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5. Memory, RAM and ROM

[4]

(a) State what is meant by volatile memory.

[1]

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(b) State one difference between RAM and ROM.

[1]

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(c) An embedded system in a washing machine

[2]

uses both RAM and ROM. State one purpose of

RAM and one purpose of ROM in this system.

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6. Types of RAM and types of ROM

[5]

(a) State two differences between SRAM and

[2]

DRAM.

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Complete the table by naming the type of ROM described in each row.

[3]

Description	ROM type
Can be programmed once by the user and cannot then be erased.	
Can be erased using ultraviolet light and then reprogrammed.	
Can be erased and reprogrammed electrically without removing the chip.	

7. Atleast 2 hours of mindful study is expected **[0.5]**

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8. Thank you for honest participation **[0.5]**

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Total = 40 marks